# L28: Common Random Numbers and Variance Reduction

## Simulation Without a Black Box

# Outline

# The Comparison Problem: Signal vs. Noise

## The Goal

Compare two system designs A and B on the same KPI (e.g., $W_q$). We want to detect a *real* design difference, not a lucky (or unlucky) random draw.

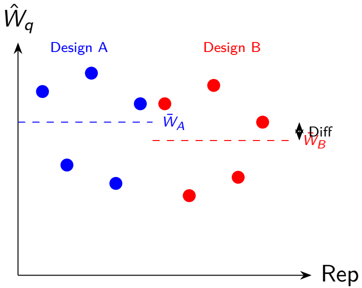
**Independent sampling approach:**

- Run $n$ replications of A with seeds $s_1, \ldots, s_n$
- Run $n$ replications of B with *different* seeds $s_{n+1}, \ldots, s_{2n}$
- Estimate $E[W_A - W_B]$ via $\bar{W}_A - \bar{W}_B$

**Variance of the difference:**

$$\mathrm{Var}(\bar{W}_A - \bar{W}_B) = \frac{\sigma_A^2}{n} + \frac{\sigma_B^2}{n}$$

This can be large when $\sigma_A^2$ and $\sigma_B^2$ are large — even if $E[W_A - W_B]$ is small.



# CRN: Same Random Streams for Both Designs

## Common Random Numbers

Use the *same* sequence of random numbers for design A and design B in each replication. Each customer arrives at the same moment and has the same inherent service demand — only the design decision differs.

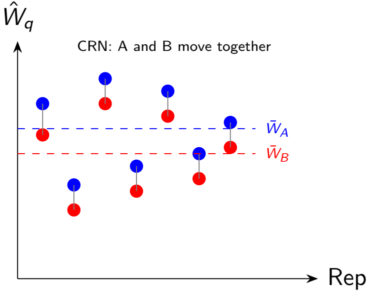
**Paired difference estimator:**

$$D_i = W_{A,i} - W_{B,i}, \qquad i = 1, \ldots, n$$

$$E[\bar{D}] = E[W_A] - E[W_B]$$

$$\mathrm{Var}(\bar{D}) = \frac{\sigma_A^2 + \sigma_B^2 - 2\,\mathrm{Cov}(W_A, W_B)}{n}$$

If $\mathrm{Cov}(W_A, W_B) > 0$: CRN **reduces variance**.



# Variance Reduction: The Math

## Variance of $\bar{D}$ Under CRN

$$\mathrm{Var}(\bar{D}) = \frac{\sigma_A^2 + \sigma_B^2 - 2\,\mathrm{Cov}(A,B)}{n}$$

Without CRN: $\mathrm{Cov}(A,B) = 0$ (independent runs).
With CRN: $\mathrm{Cov}(A,B) > 0$ if both designs respond similarly to the same random inputs.

**Variance reduction ratio (VRR):**

$$\mathrm{VRR} = 1 - \frac{\mathrm{Var}_{CRN}}{\mathrm{Var}_{ind}} = \frac{2\,\mathrm{Cov}(A,B)}{\sigma_A^2 + \sigma_B^2} = 2\,\mathrm{Corr}(A,B) \cdot \frac{\sigma_A \sigma_B}{\sigma_A^2 + \sigma_B^2}$$

If $\mathrm{Corr}(A,B) = 0.8$, $\sigma_A \approx \sigma_B$: VRR $\approx 80\%$ — CRN achieves the same precision with $\sim 5\times$ fewer replications.

## When CRN Fails

- Designs respond *oppositely* to randomness: CRN can *increase* variance ($\mathrm{Cov} < 0$)
- Happens when one design benefits from heavy load while the other is hurt by it
- Always check: is $\mathrm{Corr}(A,B) > 0$?

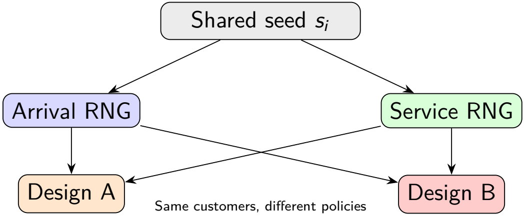
```
corr = np.corrcoef(
    results_A, results_B)[0,1]
print(f"Corr(A,B) = {corr:.3f}")
# If negative: CRN is counterproductive
```

# CRN Implementation: Stream Synchronization

**Critical requirement:** Arrival stream and service stream must be synchronized across designs A and B.

Use separate child generators for each process type so that the $k$-th customer has the same interarrival and service demand regardless of which design is running.

```python
def run_one_crn(seed, design):
    """
    seed: SeedSequence child
    design: "A" or "B"
    """
    # Spawn two sub-streams per rep
    s_arr, s_svc = seed.spawn(2)
    rng_arr = np.random.default_rng(s_arr)
    rng_svc = np.random.default_rng(s_svc)
    # Use rng_arr for interarrivals
    # Use rng_svc for service times
    # Design A/B affects # servers or
    # scheduling rule, not the RNG
    ...
```



# Paired-Difference CI and Three-Way Comparison

## Paired-Difference 95% CI

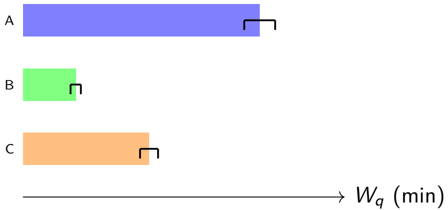
$$\bar{D} \pm t_{n-1,0.025} \cdot \frac{s_D}{\sqrt{n}}$$

where $D_i = W_{A,i} - W_{B,i}$ and $s_D$ = std of differences.

**Three-way comparison example** (30 reps, CRN):

| Design | $\bar{W}_q$ | $s$ | 95% CI |
|---|---|---|---|
| A (1 server) | 18.4 | 3.2 | [17.2, 19.6] |
| B (2 servers) | 4.1 | 0.9 | [3.7, 4.5] |
| C (1+triage) | 9.8 | 1.8 | [9.1, 10.5] |

With CRN, $s_D$ for B vs. A might be 1.5 (vs. 4.0 without CRN) — the CI is 2.7× narrower.




### Warning

With CRN, do *not* use independent-sample $t$-tests across designs. Use paired $t$-test on $D_i = W_{A,i} - W_{B,i}$.

# Key Takeaways

1. **The comparison problem**: independent runs conflate design differences with random noise.
2. **CRN** uses the same random streams for both designs in each replication, inducing positive correlation.
3. **Variance formula**: $\mathrm{Var}(\bar{D}) = (\sigma_A^2 + \sigma_B^2 - 2\,\mathrm{Cov})/n$; CRN reduces variance when $\mathrm{Cov} > 0$.
4. **Implementation**: spawn separate arrival and service sub-streams per replication; share both across designs.
5. **Paired CI**: apply the paired-difference $t$-CI; do not use the two-sample formula with CRN data.
6. **When CRN fails**: if designs respond oppositely to load, CRN can *increase* variance — always verify $\mathrm{Corr}(A, B) > 0$.

## Next Module: M09 — Experimental Design

L29 — $2^k$ factorial designs for simulation experiments.